THE CELL CYCLE AND MITOSIS (1)

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EUKARYOTIC CELLS

SOMATIC CELLS

Cells that make your organs and tissues.

SEX CELLS

The ONLY cells that are not somatic – sperm and egg (Gametes)

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CELL DIVISION

- In eukaryotes, cell division occurs in <u>two major stages</u>.
- The first stage, division of the cell nucleus, is called **mitosis.**
- The second stage, division of the cell cytoplasm, is called **cytokinesis.**

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CELL CYCLE

- The cell cycle is the series of events that cells go through as they grow and divide.
- Interphase is the period of growth that occurs between cell divisions.
- Cell spends most of its life in this phase.

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CELL CYCLE

During the cell cycle:

1. a cell grows
2. prepares for division
3. divides to form two daughter cells, each of which begins the cycle again

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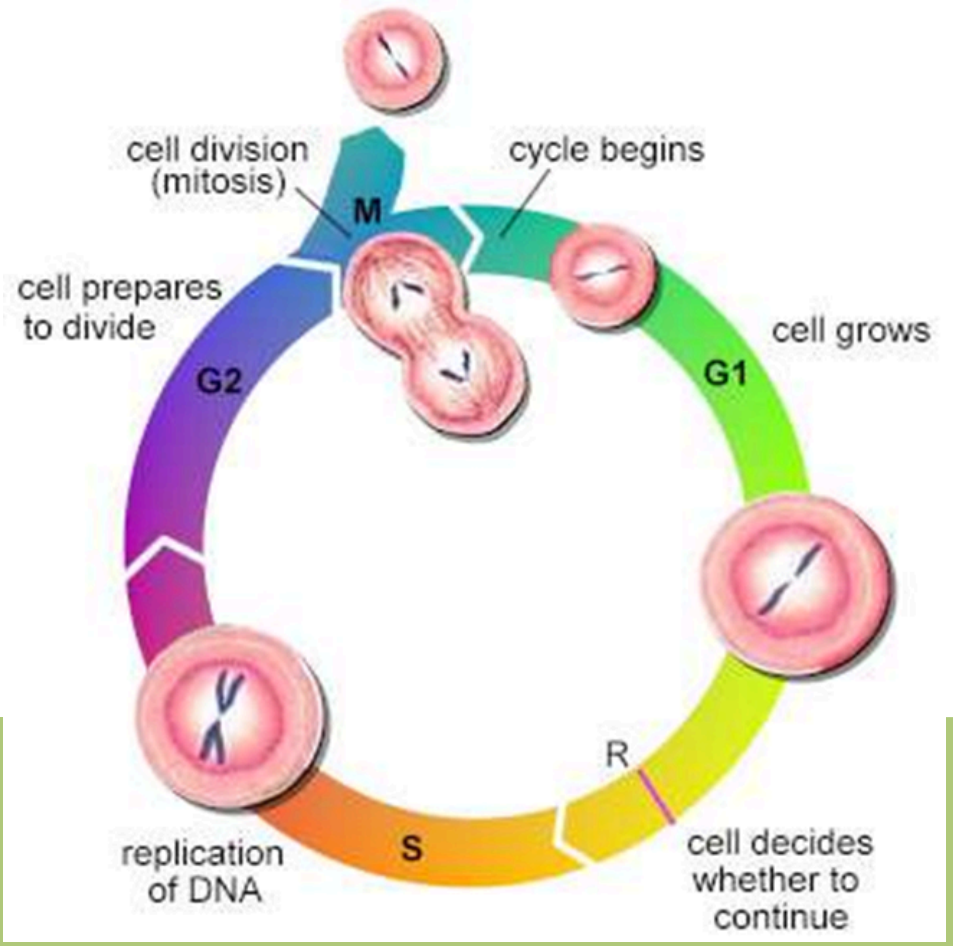
CELL CYCLE

- The cell cycle consists of four phases:
- G1 (First Gap Phase)
- S Phase
- G2 (Second Gap Phase)
- M Phase



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EVENTS OF THE CELL CYCLE

During G1, the cell:

- increases in size (cell grows)
- synthesizes new proteins and organelles (cell develops)

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EVENTS OF THE CELL CYCLE

During the S phase, (DNA replication)

- chromosomes are replicated
- DNA synthesis takes place

Once a cell enters the S phase, it usually completes the rest of the cell cycle.

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EVENTS OF THE CELL CYCLE

The G2 Phase (Second Gap Phase)

- organelles and molecules required for cell division are produced
- Once G2 is complete, the cell is ready to start the M phase—Mitosis

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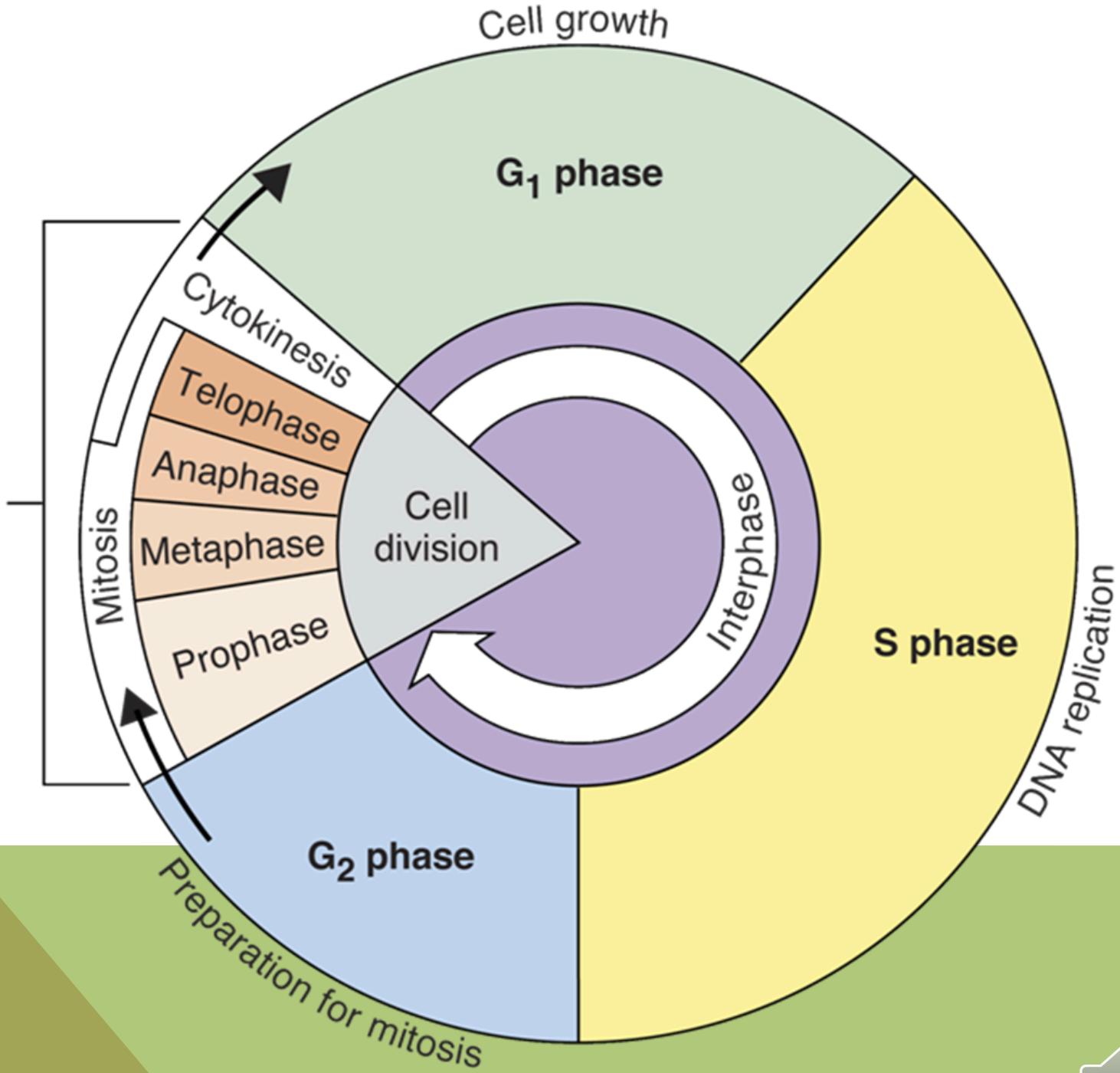
Cell Cycle



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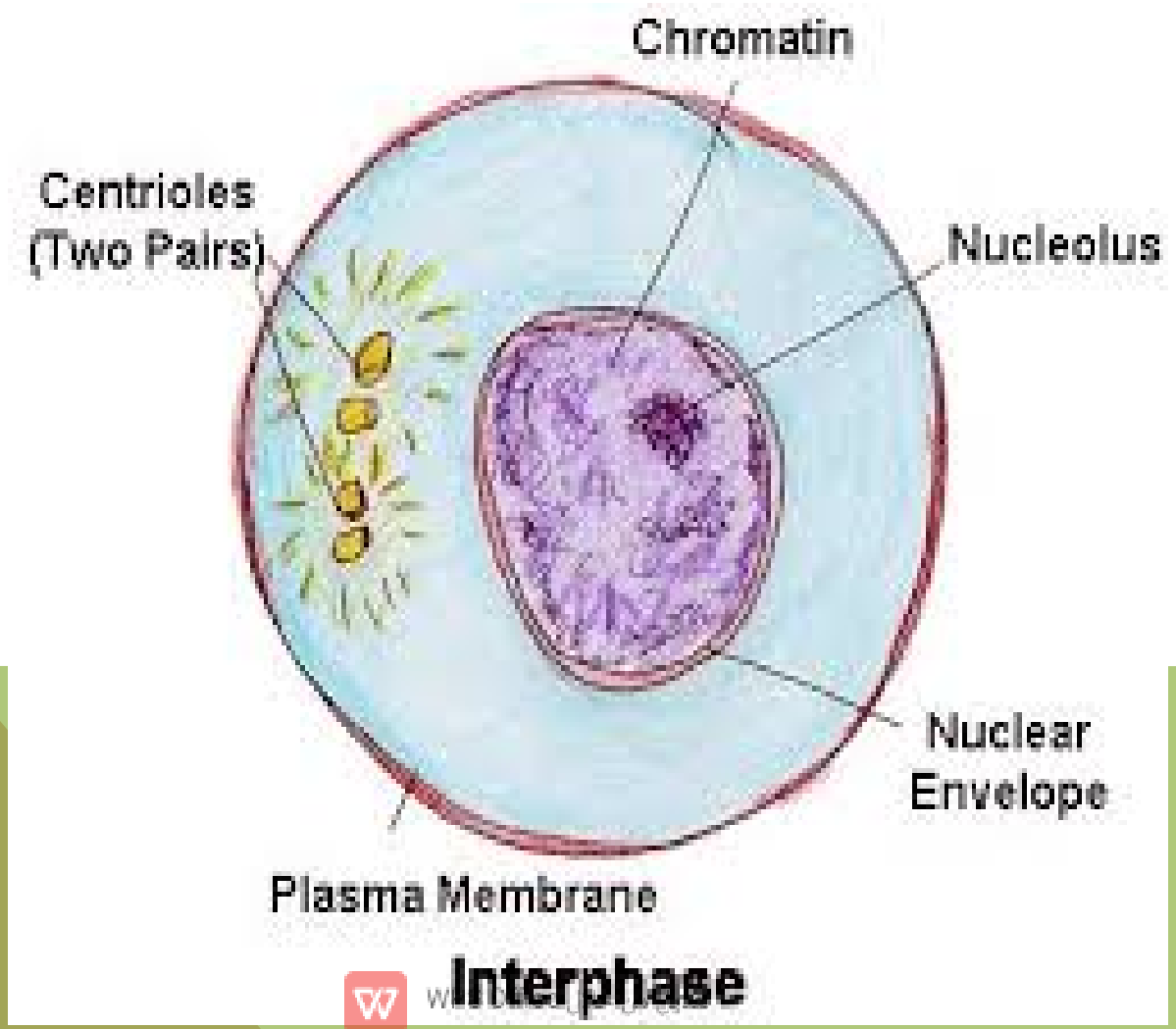
WHY DOES INTERPHASE HAPPEN?

cell must grow in order for extra organelles and DNA have room in cell



DNA

Chromatin – long, thin strands made up of DNA and protein. The protein helps the DNA to stay together when chromosomes form.

Chromosomes -

Genetic information is passed from one generation to the next on **chromosomes**.

Before cell division, each chromosome is duplicated, or copied.

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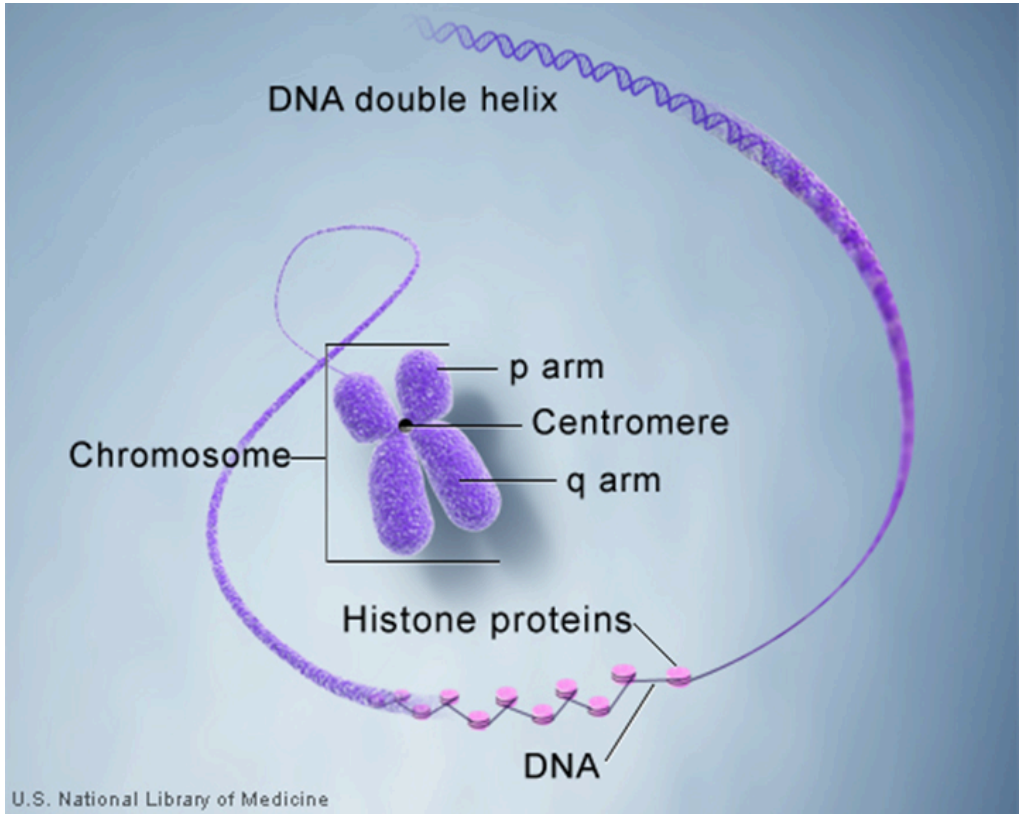
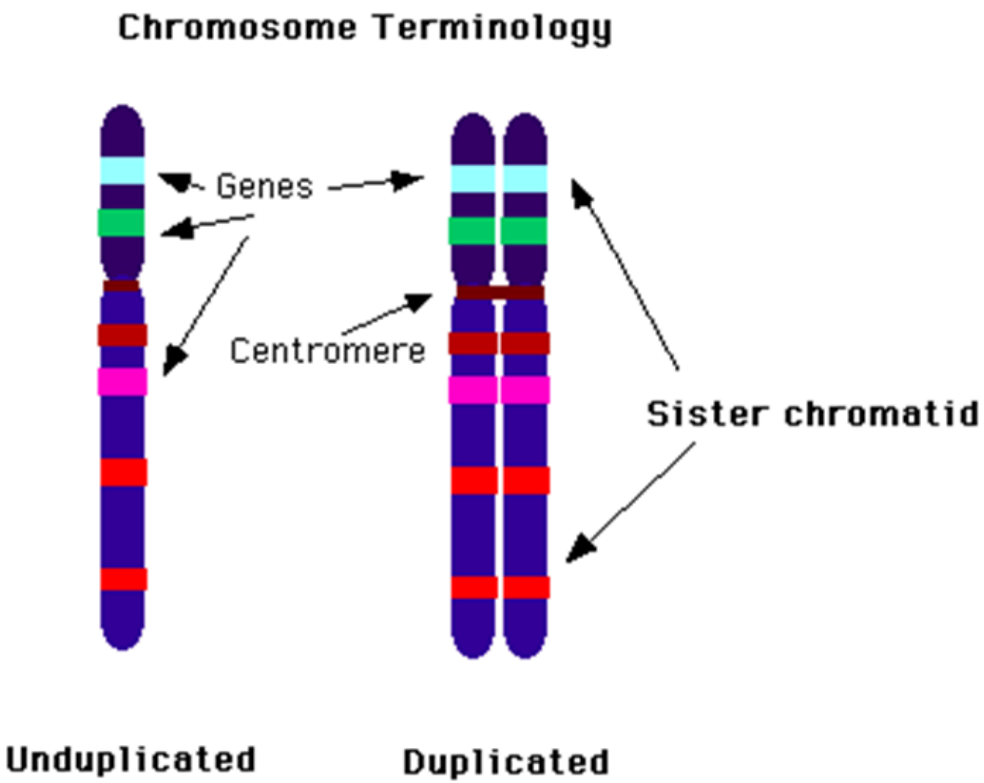
CHROMOSOMES – CONDENSED CHROMATIN





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